



ROOT CAUSE UNLOCKED · CONDITION KIT

The Post-Viral Root Cause *Kit*

What EBV serology actually means, which tests are worth running, and exactly what to ask your doctor for

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

The Post-Viral Root Cause Kit

What EBV serology actually means, which tests are worth running, and exactly what to ask your doctor for

BEFORE YOU START

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This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek emergency care now for: chest pain or pressure, new or worsening breathlessness, coughing blood, a swollen or painful calf, fainting rather than lightheadedness, new weakness or numbness on one side, trouble speaking, or severe headache of sudden onset. Blood clots are a recognized complication after COVID, and post-viral fatigue is never the safe assumption when these appear.

Also see a doctor promptly for unintentional weight loss, fever or night sweats that persist, a new lump, or fatigue that has worsened steadily rather than fluctuating.

If you have thoughts of harming yourself, call or text 988 in the United States, or go to an emergency department.

Read this before anyone prescribes you exercise

If your symptoms reliably worsen a day or two after exertion, and that worsening is out of proportion to what you did, that is **post-exertional malaise**. It is Key 6 of *Unlock Your Post-Viral Recovery* and it is the guardrail on everything else.

Where post-exertional malaise is present, graded exercise is not the answer and pushing through causes harm. This matters more in this kit than anywhere else on the shelf, because you will find exercise reconditioning recommended for one of the other things in this guide, and the two pieces of advice collide. I deal with that collision head-on in the autonomic section rather than leaving you to discover it.

The nine keys, and what tests each one

Unlock Your Post-Viral Recovery works through nine root causes. Blood work reaches some of them and not others, and being clear about which is the point of this guide.

Key	What it means	How you find it
1. Persistence	What “the infection never fully left” does and does not mean	Mostly a research question. No routine test
2. Immune activation	The inflammatory signature	hs-CRP, CBC with differential
3. EBV serology	What the antibody panel actually says	VCA IgG, VCA IgM, EBNA-1 IgG. See below, this is the most misread panel in the category
4. Reservoir	The viral reservoir hypothesis	Research setting. Not a test you can order
5. Gut and immune terrain	Absorption and the microbial contribution	Celiac serology, stool analysis where symptoms justify it
6. Mitochondria and PEM	Energy production, and the post-exertional guardrail	History, not a test. The most consequential finding here
7. Autonomic	POTS-type symptoms, orthostatic intolerance	A standing test in the office. Not a blood test
8. Mast cells and histamine	Why everything feels reactive	Baseline tryptase, with strict criteria. See below
9. Testing	Choosing the right panel rather than the biggest one	This guide

Two of the nine cannot be tested outside research, two more are answered by history or a bedside test rather than blood, and one, EBV serology, is more often misread than read. That is the honest map.

Key 3: EBV serology, the most misinterpreted panel in this category

This is the section I would keep if I could keep only one, because a great deal of money is spent on the back of a misreading.

What the panel measures. The standard EBV panel reports antibodies to viral capsid antigen, both IgM and IgG, and to Epstein-Barr nuclear antigen 1. In an evaluation of these assays for determining EBV serostatus, patients without heterophile antibodies who had **VCA IgG and EBNA IgG antibodies were supporting past infection** (PMID 28661055).

What that means for you, stated as plainly as I can.

- **VCA IgG positive plus EBNA-1 IgG positive is the signature of past infection.** It means the virus met your immune system at some point, most often in adolescence.
- **Roughly nine in ten adults carry that pattern.** It is close to universal.
- **It is therefore not a diagnosis, and it does not explain your fatigue.** If a positive EBV IgG were sufficient to explain post-viral symptoms, most of the adult population would have them.

Where the misreading happens. A panel comes back with “high” VCA IgG, the number is flagged above a reference line, and it gets presented as proof of chronic reactivated EBV driving the illness. Titre height on IgG is not a reliable measure of current viral activity, and a raised IgG in someone with past infection is an ordinary finding.

What would actually be different. Chronic active EBV disease is a genuinely recognized entity, and it is **rare and severe**, presenting with persistent fever, organ involvement, and lymphoproliferative complications rather than with fatigue alone. It is diagnosed in specialist settings, usually with viral load quantification rather than antibody titres. If a clinician genuinely suspects it, that is a haematology referral, not a supplement protocol.

So what should you do with an EBV panel? Run it once if it has never been done, mainly to establish serostatus. Do not repeat it to track progress, because it will not track anything useful, and do not let a positive IgG become the explanation that stops the search.

Key 7: the autonomic system, and the contradiction you need to know about

This is the key most often missed, and it is also where the safety advice in this field genuinely conflicts.

How common it is. A cross-sectional study of long COVID patients found dysautonomia common and contributing to overall symptom burden, concluding that identifying it has diagnostic and therapeutic implications (PMID 41951431). A systematic review examined the prevalence of orthostatic intolerance,

postural orthostatic tachycardia syndrome, and orthostatic hypotension in post-acute sequelae of COVID (PMID 41584279).

What it feels like. Lightheaded on standing, heart racing when upright, worse in heat, worse after meals, better lying down, brain fog that lifts when horizontal. If you have quietly started sitting down in showers, that is this.

How it gets found, and it is not a blood test. A standing test in the office: lie down for ten minutes, record heart rate and blood pressure, then stand and record at intervals over ten minutes. A sustained heart rate rise of 30 beats per minute or more in adults on standing, without a fall in blood pressure, is the POTS pattern. Ask for this by name. It takes fifteen minutes and costs nothing.

The contradiction, stated directly

For POTS, the literature recommends exercise reconditioning. Cardiovascular deconditioning, meaning cardiac atrophy and low blood volume, contributes significantly to POTS and its disability, so physical reconditioning with exercise training is a recommended approach, alongside measures such as increased salt and fluid intake and compression, while the authors note that randomized trials are still needed to evaluate these non-pharmacological treatments (PMID 30001836).

For post-exertional malaise, exercise is contraindicated. Pushing causes harm.

Many people in this population have both. So which wins?

The post-exertional malaise rule wins. If you have PEM, an exercise prescription designed for POTS can make you worse, and the reconditioning literature was not written with PEM in mind. This is the single most important thing in this guide to raise with a clinician, because a well-meaning graded exercise plan aimed at your POTS is a realistic way to lose six months.

What is safe to do meanwhile, and is recommended in the same non-pharmacological literature: increase salt and fluid intake if your clinician agrees it is appropriate for you, use compression garments, change position slowly, and use recumbent or horizontal movement rather than upright exertion if you are attempting any activity at all. Those address the orthostatic side without triggering the PEM side.

Key 8: mast cells and histamine, with the criteria that actually apply

Why it is here. A proportion of people in this group describe reacting to foods, smells, medications, heat, and alcohol in ways they never did before.

What has been reported. A study of long COVID patients with cardiovascular manifestations attributed to mast cell activation reported that antihistamines improved cardiovascular manifestations and other symptoms, with no significant change in controls, and the authors take this as supporting a role for mast cell activation in long COVID pathophysiology (PMID 37529714).

Read that carefully. It is a small non-randomized study, the comparison group was not a placebo arm, and “responded to antihistamines” is suggestive rather than diagnostic. It is enough to make a trial of an over-the-counter antihistamine a reasonable conversation with your clinician. It is not enough to establish a diagnosis.

And the diagnostic discipline matters here. Mast cell activation syndrome has consensus criteria requiring typical episodic symptoms in at least two organ systems, a substantial rise in serum tryptase above

the person's own baseline during an event, and response to mast-cell-directed treatment. A critical review found that the alternative criteria circulating in practice often lack specificity and validation, and that an increasing number of patients are being given a misdiagnosis on the strength of them (PMID 34166845).

Practically: a baseline serum tryptase is reasonable if reactions are systemic. A tryptase drawn during an actual event, compared against that baseline, is the informative test. The Allergy kit covers this in more depth, including hereditary alpha-tryptasemia, which explains most raised baseline tryptase results.

Keys 2 and 5: the terrain, and the ordinary blood work

Inflammation. hs-CRP and a CBC with differential. Do not expect them to be dramatic; in most post-viral presentations they are unremarkable, and a normal CRP does not mean nothing is wrong.

The gut. A case-based systematic review of brain-gut axis approaches in long COVID and ME/CFS notes that the two conditions share persistent fatigue, post-exertional malaise, and gastrointestinal dysfunction, with growing evidence implicating brain-gut axis dysregulation, while calling for larger mechanistic studies and rigorously designed randomized trials (PMID 41668172). Directional, and honest about its own limits.

Practically: celiac serology while still eating gluten, and stool analysis only if gut symptoms genuinely justify it.

The deficiencies worth excluding, because they are common, correctable, and produce exactly these symptoms: ferritin with a full iron panel, B12, RBC folate, homocysteine, vitamin D, and thyroid function with TSH, free T4 and free T3. Ferritin is the one most often waved off, because a result inside a lab's standard range still gets reported as normal. Read it against the functional window in the table below rather than against the lab's flag, and see the supplement guide in this kit for the trial evidence behind that threshold.

How common is this, and is it real

Both questions get asked, so: a systematic review and meta-analysis of prospective cohort studies found long COVID affects over one-fifth of people who had SARS-CoV-2, with fatigue and breathlessness persisting in many, and female sex and severe acute infection increasing risk (PMID 41462874). A separate systematic review and meta-analysis of observational studies emphasizes that reported prevalence varies substantially with definitions and that results should be read as a descriptive summary of heterogeneous evidence rather than a single precise rate, while calling for rehabilitation pathways and multidisciplinary care (PMID 42428921).

It is real, it is common, and the prevalence numbers are genuinely uncertain because the definitions are.

Functional ranges for the markers in this guide

Clinical targets used at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose.

Marker	Units	Functional target	Approximate standard range
Ferritin	ng/mL	50 to 150	25 to 300
Serum iron	ug/dL	65 to 110	40 to 150
TIBC	ug/dL	290 to 370	250 to 400
Transferrin saturation	%	20 to 40	15 to 45
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Homocysteine	umol/L	under 7.0	under 10.0
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
hs-CRP	mg/L	under 0.5	under 1.0
White blood cells	K/uL	5.0 to 7.5	3.5 to 10.5
Lymphocytes	%	25 to 40	15 to 45
Glucose, fasting	mg/dL	75 to 86	70 to 99
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0

Thyroid targets are deliberately not published here. I only publish ranges I can transcribe exactly from a validated source, and the right thyroid target depends on age, pregnancy, and whether you are already on replacement.

The conversation with your doctor

A request list you can bring

- CBC with differential, hs-CRP
- Ferritin, serum iron, TIBC, transferrin saturation
- Vitamin B12, RBC folate, homocysteine
- 25-hydroxy vitamin D
- TSH with free T4 and free T3
- Celiac serology, drawn while still eating gluten
- EBV panel **once**, to establish serostatus, if never done
- Baseline serum tryptase, if reactions are systemic or involve more than one organ system
- **A ten-minute standing test**, which is free and is the highest-yield item on this list

The sentence that changes the appointment

“My symptoms are worse when I am upright and better lying down. Could we do a ten-minute standing test with heart rate and blood pressure?” That is specific, cheap, and it finds the thing most often missed.

And the sentence that protects you: “My symptoms reliably get worse a day or two after exertion, so I need any activity plan to account for post-exertional malaise.”

What these labs cannot tell you

- **Whether a virus is still active in your tissue.** The reservoir question is a research question and there is no clinical test.
- **Whether your EBV is reactivated.** A positive IgG means past infection, which most adults have.
- **Whether you have post-exertional malaise.** History answers that, and it matters more than the panel.
- **How long recovery will take.** Nobody can tell you that honestly, and anyone who does is guessing.

When to seek clinical review

Emergency: chest pain, new or worsening breathlessness, coughing blood, swollen painful calf, fainting, one-sided weakness or numbness, trouble speaking, sudden severe headache.

Soon: unintentional weight loss, persistent fever or night sweats, a new lump, steadily worsening rather than fluctuating symptoms, fainting on standing, or a resting heart rate that has changed markedly.

Now: thoughts of harming yourself. Call or text 988.

Your next step

The companion guides do the rest: **The Post-Viral Diet Guide** covers the gut, histamine, and eating within an energy budget, and **The Post-Viral Supplement Guide** covers pacing, what has been studied, and what to skip.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-labs

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-labs

If you would rather start on your own, start tracking, and track exertion alongside symptoms so the delay between them becomes visible:

Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-labs

References

All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in [Post-Viral-Kit-Citation-Ledger.md](#).

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Educational content only. Not medical advice. Not a diagnosis. Not a substitute for your clinician.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek emergency care for: chest pain or pressure, new or worsening breathlessness, coughing blood, a swollen painful calf, fainting rather than lightheadedness, one-sided weakness or numbness, trouble speaking, or sudden severe headache.

If you are pregnant, nursing, underweight, diabetic, or have a history of disordered eating, talk with a clinician before restricting anything. In this population an unnecessary restriction costs more than usual, for the reason in the next section.

If you have thoughts of harming yourself, call or text 988.

The constraint that governs this whole guide

Cooking is an activity. Shopping is an activity. Standing at a counter chopping vegetables comes out of the same daily budget as showering, working, and answering messages.

Most post-viral diet advice is written as though the reader has normal energy to implement it, then prescribes an elimination protocol with three cooked meals a day. The person following it spends their entire envelope on food preparation and crashes.

So every strategy here has to cost less energy than it returns, and where something is expensive to run, I say so.

If your symptoms reliably worsen a day or two after exertion, that is post-exertional malaise, and this is not a nicety. Read the pacing section of the supplement guide first.

Where food sits among the nine keys

Unlock Your Post-Viral Recovery works through nine root causes. Food reaches three of them: the gut and immune terrain, the histamine and mast cell picture, and the nutrient side of the mitochondrial key. It does not reach viral persistence, it does not reach post-exertional malaise, and it does not fix orthostatic intolerance, though the salt and fluid part below helps that one.

Knowing which keys food can actually touch is what stops you spending six months on the wrong one. The lab guide maps all nine.

Part one: the two changes that help most people here

Salt and fluid, if standing is a problem

This is the highest-yield dietary change in this guide for anyone whose symptoms are worse upright, and it is unusual in being a dietary change with a physiological target rather than a general health goal.

Increased salt and fluid intake sits among the non-pharmacological measures described for postural orthostatic tachycardia syndrome, alongside compression, with the authors noting that randomized trials are still needed to evaluate these approaches (PMID 30001836).

How to do it. Fluid steadily across the day rather than in large boluses. Salt added deliberately to food, or an electrolyte drink without much sugar. Drink a glass of water before standing up in the morning, before showers, and before anything that reliably makes you lightheaded.

Clear this one with a clinician first. Salt loading is genuinely inappropriate in high blood pressure, heart failure, and kidney disease, and this is the one recommendation in the kit where getting it wrong for your situation matters.

Eating in a way that does not itself cause symptoms

Large meals pull blood to the gut and reliably worsen both orthostatic symptoms and post-meal crashes in this population.

- **Smaller meals, more often.** Four or five small ones beats two or three large ones here.
- **Protein at breakfast**, which steadies the rest of the day.
- **Sit down to eat**, and stay sitting for a while afterwards if standing after meals is a problem.
- **Watch the post-meal window.** If you reliably feel worse forty minutes after eating, meal size is the first thing to change, before the food list.

Part two: the histamine question, scoped honestly

Key 8 of the ebook covers mast cells and histamine, and a share of readers will recognize themselves in it: reacting to foods, alcohol, heat, and smells in ways that are new since the illness.

What has been reported. In long COVID patients with cardiovascular manifestations attributed to mast cell activation, antihistamines improved cardiovascular and other symptoms, with no significant change in controls (PMID 37529714). That is a small non-randomized study without a placebo arm, and it is enough to make the possibility worth taking seriously rather than enough to diagnose anything.

What a dietary trial looks like, and what it costs. A low-histamine trial is a real option and it is one of the more expensive things in this guide in energy terms, because the operating principle is freshness rather than a forbidden list. Meat cooked and eaten the same day is low histamine; the same meal on day three is not. That means more frequent cooking, which is exactly the resource you do not have.

So run it as a bounded, cheap version:

- **Four weeks, not indefinitely.**
- **Change the freshness variable first**, before you remove whole food groups. Cook, portion, and freeze immediately rather than refrigerating leftovers for days. For many people this is most of the available benefit and it costs no extra cooking.
- **Then remove the top few candidates** if that alone was not enough: aged cheese, cured and smoked meat, fermented foods, alcohol, and leftovers of any protein.
- **Reintroduce one at a time afterwards.** Skipping this is how a four-week trial becomes a permanently narrowed diet.
- **If four weeks changes nothing, stop.** Do not escalate to a stricter version.

The Allergy Diet Guide covers the histamine evidence in full, including the placebo-controlled challenge data, and it is worth reading before you commit to this.

Part three: the gut key, without overclaiming

Why it matters concretely. Absorption. You cannot correct an iron or B12 deficiency through a gut that is not absorbing, and those deficiencies produce exactly these symptoms.

Get celiac testing before removing gluten. The serology is only interpretable while you are still eating it. Remove it first and you lose a clean answer without a deliberate gluten challenge later.

What the research supports, and its limits. A case-based systematic review of brain-gut axis approaches in long COVID and ME/CFS notes the shared features of persistent fatigue, post-exertional malaise, and gastrointestinal dysfunction, with growing evidence implicating brain-gut axis dysregulation, while calling for larger mechanistic studies and rigorously designed randomized trials (PMID 41668172). A review of long COVID management discusses probiotics for gastrointestinal health and wellbeing and concludes that large-scale randomized controlled trials are warranted to validate their efficacy (PMID 41862909).

Read both as directional. Plausible, actively researched, not demonstrated. I am not going to tell you a gut protocol resolves post-viral illness, because no trial says so and in this population the cost of a wrong bet is measured in months.

Part four: eating within an energy budget

This is the practical heart of the guide.

Cook once, eat three times. Batch on your best day, portion immediately, freeze. Frozen portions, not five-day-old fridge leftovers, which matters doubly if histamine is part of your picture.

Keep a stocked no-cook tier at all times. Not a failure plan, a normal part of the system: tinned fish, pre-cooked rice, frozen vegetables, eggs, nut butter, fruit, plain yogurt if dairy is fine for you.

Lower the bar for what counts as a meal. Protein plus a vegetable plus a carbohydrate. It does not need assembling or photographing.

Convert money into energy where you can. Pre-chopped vegetables, frozen produce, rotisserie chicken, grocery delivery. In this condition that is the correct direction of trade.

Sit down. Every time. Standing to eat costs more than you think and it worsens post-meal orthostatic symptoms.

A rhythm rather than a menu. Batch two proteins and two vegetables on your best day, cook a grain that keeps, rotate for three days, use the no-cook tier for two, repeat.

What I deliberately left out, and why

- **Elimination diets beyond the bounded histamine trial.** Expensive to run in energy terms, they narrow nutrition, and the yield is low. Celiac testing is the exception and it is a blood test, not a diet.
- **Extended fasting.** Skipping meals works against the blood sugar and orthostatic work above, and in this population it commonly backfires.
- **Antiviral diets aimed at “chronic EBV.”** They rest on a misreading of the serology, which the lab guide covers. Roughly nine in ten adults carry the antibody pattern being treated (PMID 28661055).
- **Detox and cleanse protocols.** No outcome evidence, real energy cost.
- **Any plan requiring three cooked meals a day.** If a plan assumes energy you do not have, it is not a plan for you.

Telling you what to skip is part of what you paid for, and here it is also giving you your afternoon back.

Running a fair experiment

Weeks 1 and 2, baseline. Change nothing. Score energy out of ten, mark crash days, and note what you did the day before each crash.

Weeks 3 to 6, one change. Salt and fluid if you are orthostatic. Meal size if you crash after eating. The freshness rule if histamine is your thread.

Week 7, judge honestly. Compare against baseline. Fewer crashes is a better signal than more energy, and it moves first.

Then one more change. Never two at once.

Realistic expectations

What is realistic. Better tolerance of being upright within a couple of weeks of getting salt and fluid right. Fewer post-meal crashes from changing meal size. Genuine improvement over months if a real deficiency gets corrected.

What is not realistic. Diet resolving post-viral illness. The recent literature calls for rehabilitation pathways and multidisciplinary care precisely because no single intervention does this (PMID 42428921).

And the thing worth saying plainly. If you have been told this would resolve if you just ate better, and you have been eating well for a year, the problem is not your diet and it is not your discipline.

When to seek professional care

Emergency: chest pain, new or worsening breathlessness, coughing blood, swollen painful calf, fainting, one-sided weakness, trouble speaking, sudden severe headache.

Soon: unintentional weight loss, persistent fever or night sweats, steadily worsening rather than fluctuating symptoms, fainting on standing, a diet narrowed below roughly twenty foods, or a markedly changed resting heart rate.

Now: thoughts of harming yourself. Call or text 988.

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your labs, your history, and whether post-exertional malaise and orthostatic intolerance are part of your picture.

***Book a free Root Discovery Session: [rootcauseunlocked.com/discovery?](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-diet)
[utm_source=guide&utm_medium=ebook&utm_campaign=postviral-diet](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-diet)***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

***Get in touch: [rootcauseunlocked.com/contact?](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-diet)
[utm_source=guide&utm_medium=ebook&utm_campaign=postviral-diet](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-diet)***

If you would rather start on your own, track first, and track exertion alongside symptoms so the one-to-two-day delay becomes visible:

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Read the companion documents alongside this one. **Understanding Your Post-Viral Labs** covers what EBV serology really means and the standing test worth asking for, and **The Post-Viral Supplement Guide** covers pacing and what has actually been studied.

References

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek emergency care for: chest pain or pressure, new or worsening breathlessness, coughing blood, a swollen painful calf, fainting rather than lightheadedness, one-sided weakness or numbness, trouble speaking, or sudden severe headache.

If you have thoughts of harming yourself, call or text 988 in the United States.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

The guardrail, before anything else

If your symptoms reliably worsen a day or two after exertion, do not let anyone talk you into pushing through. That is post-exertional malaise, it is Key 6 of the ebook, and where it is present graded exercise causes harm.

This matters more here than anywhere else on this shelf, because **you will be told to exercise for your POTS.** The lab guide explains that collision in full. The short version: if you have post-exertional malaise, the PEM rule wins, and any activity plan has to be built around it.

Two foundations that outrank everything you can buy

Pacing

A scoping review of pacing in myalgic encephalomyelitis and chronic fatigue syndrome describes it as a prominent energy-management strategy while stating plainly that the literature base is not yet sufficient to dictate treatment practice (PMID 37838675). Co-production work with adults living with ME/CFS places energy management at the center of daily life alongside sleep, diet, and psychological support (PMID 39961545). Post-infectious conditions are increasingly recognized as a group needing better research and inclusive trial design (PMID 41967005).

The honest read: pacing is the consensus practical approach, its trial base is thinner than it should be, and it is the approach least likely to harm you. In a condition where the standard alternative caused harm, that last point carries real weight.

In practice:

- Plan to your **average** day, not your best one.
- Short blocks with rest between, rather than one push and a crash.
- Stop before you are empty. Finishing the task is what costs you tomorrow.
- Track exertion alongside symptoms, so the one-to-two-day delay becomes visible.

The orthostatic basics, if standing is part of your problem

If you are lightheaded upright, racing when you stand, and better lying down, these are the non-pharmacological measures described alongside exercise reconditioning in the POTS literature, and unlike the exercise part they do not conflict with pacing: increased salt and fluid intake where your clinician agrees it is appropriate, compression garments, and slow position changes (PMID 30001836). The same source notes that randomized trials are still needed to evaluate these non-pharmacological treatments, so take them as sensible and not as proven.

Salt is the one to clear with a clinician first, because it is not appropriate in high blood pressure, heart failure, or kidney disease.

Sleep, which is unrefreshing here rather than merely short

Unrefreshing sleep is a defining feature of this illness, not a side effect of it. People in this group commonly sleep eight or nine hours and wake as though they had not slept, and that is different from insomnia.

What that changes. Chasing more hours does not fix unrefreshing sleep. What helps is protecting the regularity, because an erratic schedule makes an already-broken repair process worse: a fixed wake time, light in the morning, and a wind-down that starts before you are desperate.

And where insomnia is genuinely the problem too, the treatment has real evidence. Cognitive behavioural therapy for insomnia produces effects on depressive symptoms sustained across the year following treatment in a systematic review and meta-analysis (PMID 42435604), and fully automated digital delivery has its own systematic review and meta-analysis

(PMID 42240717). The digital versions matter here specifically, because attending weekly in-person appointments is itself an exertion this population may not have.

Rule out sleep apnea if you snore, wake with headaches, or someone has seen you stop breathing. It is common, treatable, and it will not respond to anything in this guide.

The stress axis, stated carefully

There is a real finding here and it is not the one usually sold.

A 2026 meta-analysis examining the neuroendocrine signature of myalgic encephalomyelitis and chronic fatigue syndrome found evidence for a bioactive cortisol deficit alongside exaggerated feedback sensitivity, integrating neuroendocrine and psychological perspectives to describe chronic stress maladaptation in fatigue syndromes (PMID 42026257).

Read the difference. That is not “your adrenals are exhausted,” and it does not mean an adrenal support product will fix it. It is a subtle regulatory finding in a diagnosed population, from a meta-analysis, and what follows from it practically is unglamorous: the stress axis is genuinely involved, so load management is part of treatment rather than a lifestyle suggestion. I do use four-point diurnal cortisol testing clinically, for the shape of the curve rather than to diagnose a named adrenal condition, which is a distinction worth holding onto in a market that blurs it.

In practice, and none of it costs anything: five to ten minutes daily of slow breathing with a longer exhale than inhale, a protected sleep window, and pacing. Being disbelieved for months is its own load, and naming that with someone competent is part of the work rather than separate from it.

How to read this guide

Test before you supplement, wherever a test exists. In this population, correcting a measured deficiency is the intervention most likely to do something.

Give it eight to twelve weeks, and track energy out of ten plus crash days.

One change at a time. Symptoms fluctuate on their own here more than in almost any condition, and a stack teaches you nothing.

Start low. Sensitivity to supplements is common in this group, and the usual starting dose is often too much. Half doses for the first week are reasonable.

Tier 1: correct what is measurably low

The lab guide covers what to test. The principle is the one I apply to every fatigue picture: iron, B12, folate, vitamin D, and thyroid status are common, correctable, and produce exactly these symptoms.

Iron deserves the specific mention. In a double blind randomised placebo controlled trial of non-anaemic women with unexplained fatigue, iron improved fatigue and the benefit was confined to those with **ferritin at or below 50** (PMID 12763985), a threshold inside most labs' normal range. A 2026 randomised placebo-controlled trial in non-anaemic iron-deficient women found the same direction (PMID 41736325).

Correct what is low, retest, and do not supplement speculatively expecting an energy effect.

Tier 2: studied in this population, with limits

Coenzyme Q10, with or without NADH

What the research shows. A prospective, randomized, double-blind, placebo-controlled trial of CoQ10 plus NADH in myalgic encephalomyelitis and chronic fatigue syndrome reported benefit for perceived cognitive fatigue and health-related quality of life, describing the combination as generally safe while calling for further work to confirm the benefit and clarify mechanism (PMID 34444817). A randomized, double-blinded, placebo-controlled study of ubiquinol-10 at 150 mg daily for 12 weeks in chronic fatigue syndrome found improvement in several symptoms including autonomic and cognitive measures (PMID 27125909).

How to read it. Two small placebo-controlled trials in an overlapping population, both positive, neither large. The population is ME/CFS rather than post-viral specifically, which is a real caveat and also the closest match available.

Typical studied range. 150 mg daily of ubiquinol-10 for 12 weeks (PMID 27125909).

Cautions. CoQ10 can reduce the effect of warfarin. It may modestly lower blood pressure, which matters if orthostatic symptoms are part of your picture. Not established in pregnancy.

Antihistamines, which are not supplements but belong here

Why they are in this guide. Because a lot of readers of this kit are quietly reacting to everything and have not connected it.

What was reported. In a study of long COVID patients with cardiovascular manifestations attributed to mast cell activation, antihistamines improved cardiovascular manifestations and other symptoms, with no significant change in the control group, which the authors take as supporting a role for mast cell activation in long COVID (PMID 37529714).

The limits, stated clearly. Small, not randomized, no placebo arm. That is enough to make an over-the-counter antihistamine trial a reasonable conversation with your clinician if reactivity is prominent. It is not enough to diagnose mast cell activation syndrome, and the consensus criteria for that diagnosis are strict, with a critical review finding that the alternative criteria in circulation often lack specificity and validation and are producing misdiagnosis (PMID 34166845).

Probiotics and the gut, honestly graded

A review of pharmacological and non-pharmacological management of long COVID discusses probiotics among approaches aimed at gastrointestinal health and overall wellbeing, and concludes that **large-scale randomized controlled trials are warranted to validate probiotic efficacy in long COVID** (PMID 41862909). A case-based systematic review of brain-gut axis approaches in long COVID and ME/CFS notes shared features including post-exertional malaise and gastrointestinal dysfunction, and calls for larger mechanistic studies and rigorously designed trials (PMID 41668172).

Translation: plausible, actively researched, not yet demonstrated. Reasonable to try if gut symptoms are prominent. Not a core recommendation.

Tier 3: ask your prescriber first

- **Salt loading**, which is genuinely useful in orthostatic intolerance and genuinely inappropriate in hypertension, heart failure, and kidney disease.
- **Anything stimulant-like.** In a population where overexertion causes crashes, a product that makes you feel able to exceed your envelope is not a benefit.
- **High-dose antiviral botanical protocols marketed for EBV.** The lab guide explains why a positive EBV IgG is not the diagnosis these protocols assume.
- **Anything at all in pregnancy or nursing.**

What I deliberately left out, and why

- **Antiviral supplement protocols aimed at “chronic EBV.”** They rest on a misreading of the serology. Roughly nine in ten adults carry the antibody pattern being treated, and a positive VCA IgG with EBNA-1 IgG indicates past infection (PMID 28661055). Genuine chronic active EBV disease is rare, severe, and diagnosed in specialist settings.
- **Repeat EBV titres to track progress.** They will not track anything useful and they cost money every time.
- **High-dose immune “boosters.”** In a condition characterized partly by immune dysregulation, more stimulation is not obviously the goal, and there is no outcome evidence.
- **Graded exercise programs sold as recovery protocols.** Where post-exertional malaise is present this is the intervention most likely to set you back.
- **Anything promising a timeline.** Nobody can honestly tell you how long this takes.

Telling you what to skip is part of what you paid for, and here it also protects your energy budget.

Running a fair trial

Weeks 1 to 4, baseline. Change nothing. Score energy out of ten, note crash days, and record what you did the day before each crash.

Weeks 5 to 16, one intervention, at half the usual starting dose for the first week.

Week 17, judge it. In this condition, **fewer crashes is a better outcome signal than more energy**, and it usually moves first.

Bring the tracker. Months of daily scores is what makes a clinician useful to you.

Realistic expectations

What is realistic. Correcting a real deficiency helps over two to three months. Pacing reduces crash frequency. Orthostatic measures make being upright more tolerable. Those are real gains.

What is not realistic. A supplement resolving post-viral illness. There is no product that does that, and the honest state of the field is that treatment is not settled, which is why the recent literature calls for rehabilitation pathways and multidisciplinary care (PMID 42428921).

Timeline. Judge over months. Expect a non-linear course with good and bad stretches.

Cost sense. If something has done nothing after twelve weeks, stop paying for it.

When to seek professional care

Emergency: chest pain, new or worsening breathlessness, coughing blood, swollen painful calf, fainting, one-sided weakness, trouble speaking, sudden severe headache.

Soon: unintentional weight loss, persistent fever or night sweats, steadily worsening rather than fluctuating symptoms, fainting on standing, or a markedly changed resting heart rate.

Now: thoughts of harming yourself. Call or text 988.

Where to find quality versions of these

Nothing here names a brand, on purpose.

Look for third-party testing. NSF, USP Verified, or equivalent.

Match the form to the trial. Ubiquinol-10 at 150 mg is what was studied.

Read the whole panel. Fatigue and “immune” blends frequently hide stimulants, which is the last thing you want here.

If you would rather not evaluate all of that yourself, I keep a curated dispensary of professional-grade products at Root Health, screened for the standards above.

Root Health dispensary: us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the highest-value items in this guide, pacing and the standing test, cost nothing at all.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your history, and two questions that change everything: whether post-exertional malaise is present, and what happens to your heart rate when you stand.

***Book a free Root Discovery Session: [rootcauseunlocked.com/discovery?](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-supplement)
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Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

***Get in touch: [rootcauseunlocked.com/contact?](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-supplement)
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If you would rather start on your own, track first:

***Free symptom tracker: [rootcauseunlocked.com/tracker?](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-supplement)
[utm_source=guide&utm_medium=ebook&utm_campaign=postviral-supplement](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=postviral-supplement)***

Read the companion documents alongside this one. **Understanding Your Post-Viral Labs** covers what EBV serology actually means and the standing test worth asking for, and **The Post-Viral Diet Guide** covers eating within an energy budget.

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Dr. Adam Seay, DC · Seay Wellness / Root Health, Garner, NC
rootcauseunlocked.com · (919) 662-0520

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